

TrES-5b

R-2592

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_260723 · created 2026-08-27T23:24:32+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461244.7587 +/- 0.0045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.084 +/- 0.038
Transit depth (Rp/Rs)^2	0.71 +/- 0.64 [%]
Orbital Inclination (inc)	89.56 +/- 2.94
Airmass coefficient 1 (a1)	0.973 +/- 0.027
Airmass coefficient 2 (a2)	0.022 +/- 0.073
Scatter in the residuals of the lightcurve fit is	2.79 %
Best Comparison Star	#3 - [138, 308]
Optimal Method	PSF photometry
Transit Duration (day)	0.0794 +/- 0.006

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.084 ± 0.038 vs 0.142 (deviation 1.47σ)
Duration fit vs published	0.0794 d vs 0.0758 d (deviation 0.57σ)
Mid-time O–C	-11.1 min
Depth SNR	2.1
Residual scatter	2.79 %

Light curve

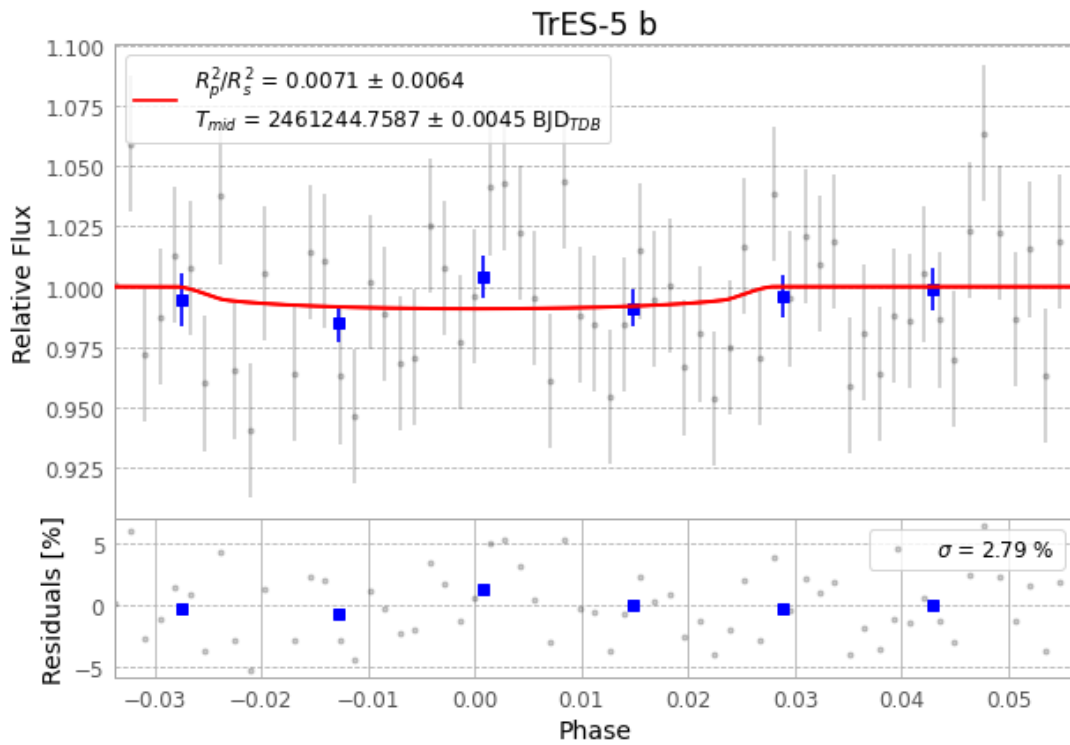


Figure 1 — FinalLightCurve_TrES-5 b_2026-07-22.png (SHA-256 b363482abb9f6df3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [234, 275], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 75d87533d56f3f53...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

64 FITS frames (42 MB), TRES-5260723045715.FITS → TRES-5260723080915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-260723.log (b74b28ebd84e...), FinalLightCurve_TrES-5 b_2026-07-22.png (b363482abb9f...), FinalLightCurve_TrES-5 b_2026-07-22.csv (7e49a2edac60...)

Compute resources

Wall time 123.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-27 23:24Z.